Final Build Report & API Cost Analysis

BK-Ingest — AI-Powered Catalog Onboarding, Proof of Concept

Prepared by	ADLM Studio — Adedolapo Quasim (engineer of record)	**Date**	18 July 2026
Client	BuildersKonnect (BK)	**Build window**	17–18 Jul 2026 · all 5 milestones complete
Repository	github.com/adedola12/Bk-PoC	**Live**	bk.adlmstudio.com (client) · api-bk.adlmstudio.com (API)
Governing docs	Knowledge Base v1.1 · Design Document v1.0 · Addendum D8–D11 · Acceptance Tests v1.0 (ratified) · BUILD_PROMPT.md · DECISIONS.md (D12–D19)	**Evaluation day**	31 July 2026 (13 days ahead of plan)

1. Executive Summary

BK-Ingest is complete and live. The system accepts arbitrary vendor uploads — PDF catalogs, datasheets, Excel, Word, product images, or a manufacturer weblink — and converts them into BK's exact bulk-upload Excel template, with AI performing the reading and a per-field confidence system deciding what passes automatically and what a human must see. Its two governing promises are demonstrated and under automated test: **nothing wrong enters the catalog silently, and nothing is dismissed silently**. Against the ratified acceptance tests, the PoC passes seven of eight outright; the eighth (field accuracy, T2) stands at 94.7% interim — 100% after brand resolution — and awaits only ~30 minutes of human ground-truth verification to become final. Every metric in this report is regenerable from the repository (`npm test · npm run eval · npm run probe`). Section 8 provides the Anthropic API cost analysis requested: measured unit economics, projections from pilot to full production, and the funding recommendation to run the system at maximum power.

2. Delivery vs. Commitment

The client brief asked for a designed solution and, where necessary, a proof of concept demonstrating AI-driven extraction, classification, mapping and transformation into the bulk-upload template with minimal human intervention — evaluated on thought process, engineering approach, scalability, accuracy and real-world handling. ADLM committed a working demonstration on the five representative catalogs within two weeks of receiving materials. Materials arrived 17 July; the complete system — pipeline, 7-page showcase UI, evaluation harness, and live deployment — was finished 18 July, thirteen days ahead of the committed window. All work remained inside PoC boundaries: standalone build, no BK codebase integration (per brief), evaluated solely on the provided sample set.

3. What the System Does, End to End

Stage	Behavior delivered
Upload & triage (D7, D12, D16)	Files dragged in or a weblink pasted (bounded crawl). Every source classified before extraction: bk_template / catalogue / product_datasheet / product_image / vendor_document / other.
Triage verification (D11)	Non-product classifications surface a snapshot + AI-summary card; a human confirms or reclassifies before filing. Confirmed vendor documents parse into structured vendor profiles (Satkay Limited is the reference case: RC 196060, 10-yr warranty, MOQ 1, 48-hr delivery).

Stage	Behavior delivered
Extraction → IPR (Stages 1–2)	One Intermediate Product Record per product; every field carries value, unit, raw string, source reference (file/page/region), confidence 0–1, and method. Raw source text is preserved verbatim forever.
Hostile-source path	Deterministic label-value pairing QA inspects each PDF page's text geometry; scrambling pages (Bosch QRG) are rasterized, panel-sliced and read by Claude vision. The 104-pp handbook runs cheap parallel text chunks with cross-chunk part-number dedupe.
Normalization (Stage 3)	Locale-aware numbers (13,45 kg → 13.45), ranges kept as ranges, pack quantities parsed, units canonicalized, per-class value-sanity rules. Violations drop confidence and flag — never fix.
Classification (Stage 4, D1/D14/D15)	Three methods in cost order (deterministic section headers → lexical/fuzzy → semantic). Agreement raises confidence; disagreement routes to review under ratified precedence rules.
Brand resolution (Stage 5, D2)	Registry with canonical names, aliases (Jaguar→Jaquar), lineage (Artize ⊂ Jaquar), product-code signatures — a QUA- code resolves to Artize even where the sheet says Jaquar; the sheet's wording stays in raw. Unknown brands auto-register as candidates with evidence.
Variants (Stage 6, D5)	Declared finish options expand to one row per purchasable variant with manufacturer finish codes substituted into product codes, labelled derived_unverified throughout.
Category IDs (D9)	ADLM-issued cat_ IDs in BK's observed format, crosswalked to the four confirmed BK IDs, provenance-labelled; missing BK IDs become todo entries — never guessed.
Content generation (Stage 8, D3)	~100–130-word descriptions + 8–9 tags in the Twyford house style (style examples read from the golden file at runtime). A deterministic fact-checker verifies every number, brand mention and standards reference against extracted data; the checker provably rejects poisoned copy (self-test in harness).
Mapping, media & gate (Stages 7, 9)	Template-driven column mapping; dropdown matching exact → synonym → recorded-custom; images pair by filename tokens incl. brand aliases, upload to Cloudinary, URLs ride the template's media columns. Per-row disposition PASS / REVIEW / TODO from a single threshold config (D18 for derived fields).
Emission & pricing (Stage 10, D4/D6/D10)	Emitter reads the BK template's own headers, dropdowns and Instructions at runtime — schema never hardcoded — and shards at the 500-row cap. Twyford golden file round-trips losslessly (21×30, zero diffs). Prices are append-only events (updates rejected at the model layer): full history per SKU, cross-brand comparison.

4. Acceptance-Test Scorecard (ratified T1–T8)

Test	Ratified threshold	Result	Status
T1 Document triage	10/10 sample files	10/10	**PASS**
T2 Field accuracy	≥95% clean · ≥85% hostile	94.7% interim; 100% after Stage-5 brand resolution	**INTERIM — pending human ground-truth verification (~30 min)**
T3 Zero-touch rate	≥70% overall	100% (Twyford 100 · Alca 100 · Jaquar 100 · Artize 100)	**PASS**
T4 Safety — no silent errors	0 leaks	A1 “2700 V” caught · A3 part no. 0601513000-on-3-products caught · 10/10 seeded errors flagged · QRG text path rejected by pairing QA	**PASS**
T5 Pricing policy	100% appended, no overwrites	Append-only enforced at model layer (proven); history queryable; 2 brands compared	**PASS**
T6 Variant policy	100% declared options	Jaquar base+9 · Artize base+9 · all derived_unverified	**PASS**
T7 Throughput	104-pp handbook < 1 hr	2.0 minutes (89 products, 52 calls)	**PASS**
T8 No silent dismissal	100% — SG_BK.docx case	100%; every non-product route surfaces the D11 prompt	**PASS**

Supporting numbers: final emission 42/42 rows at 100% zero-touch · 21 generated descriptions, 0 fact-check failures (1 auto-regeneration) · 57/57 automated tests green · vision probe ≈ 7.1 s / $0.008 per page. T2 is reported as interim, not passed: under the ratified strategy, accuracy is only final once measured against human-verified ground truth (annotate-first rule, strategy §4.1). The interim 94.7% is retained in the record; its sole miss (Artize ⊂ Jaquar lineage) was resolved exactly where the design predicted — Stage 5.

5. Architecture, Stack & Interface

Stack (D8, D13): React 19 + Vite + Tailwind v4 client (BK navy/gold, Plus Jakarta Sans); Node ≥20 ESM / Express 5 / Mongoose 8 server with stage-per-module pipeline, template-driven emitter, taxonomy + ID register, and the T1–T8 evaluation harness. MongoDB Atlas with nine collections (UploadLedger, IPR, BrandRegistry, TaxonomyNode, IdCrosswalk, PriceEvent [append-only], ReviewItem [reviewer resolutions kept as calibration data], TodoItem, Vendor). Cloudinary for media. exceljs, pdfjs-dist + pdf-to-img + sharp, mammoth for formats. Claude (claude-sonnet-4-6) for triage, extraction, vision, semantic classification and generation — structured JSON, retries with backoff, per-call cost and latency logged to every run record. Deployed: client on Vercel, API on Render, auto-deploy from GitHub main. All run artifacts write to runs/<timestamp>/; fixtures/ is read-only.

Interface (7 pages): Upload (drag-drop + weblink) · Triage Verify (D11 cards) · Review Queue (accept / correct / reject, corrections retained for calibration) · Todo (price-missing with inline entry; BK-ID-pending) · Products (card grid, Cloudinary covers, disposition & derived badges, one-click extract and emit with xlsx download) · Price Compare (per-SKU history, cross-brand) · Run Report (live T1–T8 scorecard).

6. Decision Trail (D1–D19)

The build ran on a ratified decision log with unbroken numbering from project chat into the repo: D1 trade taxonomy + 3-method classification · D2 dynamic brand registry · D3 facts-only generated content + fact-check · D4 price → blank + todo + delta re-upload · D5 one row per variant, derived SKUs labelled · D6 the template IS the contract · D7 triage first · D8 MERN / Anthropic / Cloudinary · D9 ADLM cat_ ID register +

crosswalk · D10 append-only price ledger · D11 no silent dismissal + vendor profiles · D12 bk_template triage label · D13 ADLM house repo layout · D14 classifier disagreement precedence · D15 semantic tier as Claude matching (PoC) · D16 weblink ingestion with bounded crawl · D17 category-profile exemptions (proposed) · D18 derived-field gate threshold (proposed) · D19 cross-file SKU reconciliation (open, backlog). D17/D18 remain formally unratified; neither blocks evaluation.

7. Known Limitations & Backlog (stated, not hidden)

#	Item	Class
1	T2 finalization requires ~30 min human verification of ground-truth drafts in ground-truth/	Pre-evaluation task
2	Fully client-rendered storefronts (JS-only catalogs) receive an honest guidance card; headless-browser fetcher is a production item	Production backlog
3	D19: re-uploading the same file should delta-update via the Upload Ledger identity index rather than create duplicate rows	Production backlog
4	Only the Wash Basins BK template exists; other categories emit through mapping profiles with visible D17 exemptions until BK issues per-category templates	Client dependency
5	Anthropic API credits must be funded for AI runs; on exhaustion the system degrades honestly — uploads route to human triage with the reason shown	Operations (see §8)

8. Anthropic API Cost Analysis — Running at Maximum Power

8.1 Current pricing (verified 18-Jul-2026)

The Anthropic API is usage-based: no subscription, prepaid credits from $5, billed per million tokens (MTok) with input and output metered separately. Rates relevant to BK-Ingest:

Model	Input / MTok	Output / MTok	Role in BK-Ingest
Claude Sonnet 4.6 (deployed)	$3.00	$15.00	All AI stages today: triage, extraction, vision, semantic classification, generation
Claude Haiku 4.5	$1.00	$5.00	Optimization candidate for triage + lexical-tier assists (3–5× cheaper)
Claude Opus 4.8	$5.00	$25.00	Not required — Sonnet meets all accuracy gates

Cost levers: **Batch API = 50% off both input and output** (Sonnet effectively $1.50/$7.50) for non-interactive jobs; **prompt caching = 90% off cached input reads** (cache writes 1.25× for 5-min TTL, 2× for 1-hr TTL) — the two stack. Rate limits scale with Anthropic usage tier (tier rises with cumulative spend); the Batch API also relieves rate-limit pressure for bulk catalog runs. Source: Anthropic pricing documentation via docs.claude.com. Third-party reports of newer models at introductory pricing exist but are unverified here; re-check rates at production cutover.

8.2 Measured unit economics (from the PoC's own run records)

Operation	Measured / derived	Basis
Vision extraction, per PDF page	≈ **$0.008 · 7.1 s**	Measured — Milestone A probe (HD_BOOKET sample)
Full 104-pp catalog, worst-case all-vision	≈ **$0.83 ·** ≈ **12 min**	Derived from measured probe
Full 104-pp catalog, actual mixed path	**89 products · 52 calls · 2.0 min**	Measured — T7 run (text path + dedupe)
Whole-pipeline cost per product (extraction → classification → generation), text path, Sonnet, no optimization	≈ **$0.02–0.05**	Estimate from per-call logs; conservative upper band
Full 10-file sample-set evaluation run	**< $5**	Estimate from run records; includes QRG vision

Figures marked Measured come from the pipeline's per-call cost/latency logs; figures marked Estimate/Derived are labelled as such and use the conservative end of observed ranges.

8.3 Scale projections — what “maximum power” costs

Scenario	Monthly volume	Sonnet, no optimization	Optimized (Batch + caching + Haiku triage)
Evaluation day + demo week	Full eval runs + live demos	$10–25 total	n/a — keep interactive
Pilot: Satkay-scale onboarding	10 vendors · ~50 catalogs · ~5,000 products	$150–250	$60–110
Growth	50 vendors · ~20,000 products	$600–1,000	$240–450
Maximum power (full supplier base)	100+ vendors · ~50,000 products	$1,500–2,500	$550–950

Projection basis: $0.02–0.05/product whole-pipeline band from §8.2, plus vision surcharge only on hostile-layout pages (a minority of real catalogs; the QRG is the stress case, not the norm). Optimized column assumes: bulk catalog runs moved to the Batch API (–50%), cached system prompts and style examples (–90% on the repeated input share), and Haiku 4.5 handling triage and classification assists (–66% on those calls) — combined, a 55–65% reduction, consistent with the levers' published arithmetic. These are planning estimates, not quotes; actuals are visible per-run in the system's own cost logs, which is the governance mechanism: BK-Ingest reports what each catalog actually cost.

8.4 Funding recommendation

When	Action	Amount
Now (credits exhausted 18-Jul)	Top up prepaid credits to cover final eval runs, ground-truth re-checks, demo rehearsals and evaluation day with full headroom	**$25–50**
Pilot month	Fund one month of Satkay-scale onboarding; monitor per-run cost logs weekly	**$100–250**
Production cutover	Move bulk runs to Batch API, enable prompt caching, introduce Haiku tier; set a monthly budget alert in the Anthropic Console; re-verify rates at cutover	**$550–950/mo at full scale**

Perspective for the commercial case: at the optimized production band, AI cost per onboarded product is roughly $0.01–0.02 — versus minutes of skilled manual data entry per product. The pipeline's cost-tiering (deterministic → lexical → semantic; vision only when pairing QA demands it) is what keeps the ceiling low, and it improves further as the registries compound and review volume falls.

9. Evaluation-Day Runbook (31-Jul-2026)

1. Top up Anthropic credits; confirm the key on Render. 2. Human-verify ground truth; flip statuses to human_verified (finalizes T2). 3. Run `npm run eval -- --full` for the final scorecard including full-handbook T7. 4. Warm the Render free-tier instance ~2 minutes before the demo. 5. Demo arc: upload mixed files → SG_BK D11 card → confirm → vendor profile · QRG forced to vision → Review Queue holding “2700 V” and 0601513000 · Products → Emit → download the BK template file · Todo → price entry → Price Compare · Run Report scorecard. 6. State T7 timing claims from local hardware, where the 2-minute figure was recorded.

10. After the PoC

Per the original engagement framing, a passed PoC moves to a commercial phase to productionize the pipeline for BK’s full supplier base: BK import-API integration (repository access to be discussed), D19 re-upload reconciliation, headless-browser weblink ingestion, per-category BK templates, the Batch/caching/Haiku cost optimizations of §8.3, and hardening of the review UI. The registries, decision log, evaluation harness and this report transfer as-is.

Prepared 18-Jul-2026 by ADLM Studio with Claude (Anthropic) as build assistant. Every metric herein is regenerable from the repository: npm test · npm run eval · npm run probe.